VENKATA BHARGAV DEVARAMPATI

Backend Software Engineer • Python • Django • FastAPI • RESTful APIs
+91-8106482936 | bhargavdevarampati24@gmail.com | GitHub | LinkedIn | Portfolio

Profile

Backend Software Engineer with 2+ years of full-time experience building production Django/DRF systems, including a real-time IoT platform for 500+ devices. Reduced page load times by 30% and reached 85% unit-test coverage. Now extending into FastAPI, async SQLAlchemy and WebSockets.

Work Experience

Invendis Technologies India Pvt. Ltd. **Bengaluru, Karnataka**
Software Engineer *May 2025 – Present*

- Developed and maintain the backend of the company's IoT monitoring platform (Django/DRF, MySQL, Celery/Redis, MQTT), handling telemetry from 500+ devices with API latency under 200 ms.
- Building remote firmware upgrades (OTA) for IoT devices, supporting single-device and bulk rollouts with Celery-based job queues and per-device status tracking.
- Replaced HTTP polling with full-duplex WebSockets, giving users live upgrade progress and notifications until each task completes.
- Implemented rate limiting and concurrency controls for time-, network- and CPU-heavy tasks such as bulk upgrades, protecting the API and device network under load.
- Optimised MySQL schemas and added Celery/Redis pipelines, boosting query performance by 40% and API responsiveness by ~35%.
- Implemented MQTT messaging and RBAC with 3+ permission tiers, improving data reliability and securing multi-user access.

Invendis Technologies India Pvt. Ltd. **Bengaluru, Karnataka**
Associate Software Engineer *Jul 2024 – May 2025*

- Handled design, implementation, testing and deployment of core Django backend modules.
- Designed multi-tenant MySQL schemas for 3 client environments and wrote unit tests reaching 85% coverage, reducing regressions by ~25%.
- Automated OpenVPN certificate provisioning with Python/Bash, cutting new environment setup time from 2 hours to 20 minutes.

VCube Technologies **Hyderabad, Telangana**
Full-Stack Web Development Intern (Python/Django) *Jul 2023 – Feb 2024*

- Worked as a backend developer on a student/employee management system built with Django and React, following Agile and testing practices.
- Built signup/login authentication, attendance record maintenance and marks-list management modules.

Projects

Real-Time Chat Application GitHub | Live Demo
FastAPI, WebSockets, SQLAlchemy, Alembic, PostgreSQL, React

- Built an async chat backend with FastAPI and WebSockets for real-time message delivery, backed by a React frontend.
- Designed REST endpoints with JWT-based authentication and permission-based authorization.
- Modelled the data layer with SQLAlchemy and Alembic migrations; migrated from SQLite to PostgreSQL with connection pooling.
- Deployed the frontend on Vercel and the API on Render, with a health-check endpoint for uptime monitoring.

Skills

Languages: Python, SQL, JavaScript, C, C++, HTML, CSS
Frameworks: Django, Django REST Framework, FastAPI, SQLAlchemy, Alembic, WebSockets, Channels
Databases: MySQL, PostgreSQL, SQLite
Messaging & Async: Redis, MQTT, Celery
API & Architecture: RESTful API design, JWT auth, RBAC, pagination, serialization, system design fundamentals
Tools & Practices: Linux, Git, Postman, Render, Vercel, Agile, code reviews, unit testing, API docs

Education

Pace Institute of Technology & Sciences **Ongole, Andhra Pradesh**
B.Tech in Computer Engineering – CGPA: 8.8/10 *Jun 2019 – May 2023*

Certifications

Python Data Structures (University of Michigan, Coursera) | Advanced Python Programming | Python Basic & Intermediate (HackerRank) | API Beginner Path (Postman Academy)